



Deep-sea sediment metagenome from Bay of Bengal reveals distinct microbial diversity and functional significance[☆]

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ARTICLE INFO

Keywords:

Bay of Bengal (BoB)
Deep-sea sediment
Shotgun metagenome
Bacterial diversity
Metabolic potential
Bioremediation genes
Antibiotic resistance genes

ABSTRACT

Bay of Bengal (BoB) has immense significance with respect to ecological diversity and natural resources. Studies on microbial profiling and their functional significance at sediment level of BoB remain poorly represented. Herein, we describe the microbial diversity and metabolic potentials of BOB deep-sea sediment samples by subjecting the metagenomes to Nanopore sequencing. Taxonomic diversity ascertained at various levels revealed that bacteria belonging to phylum Proteobacteria predominantly represented in sediment samples NIOT_S7 and NIOT_S9. A comparative study with 16S datasets from similar ecological sites revealed depth as a crucial factor in determining taxonomic diversity. KEGG annotation indicated that bacterial communities possess sequence reads corresponding to carbon dioxide fixation, sulfur, nitrogen metabolism, but at varying levels. Additionally, gene sequences related to bioremediation of dyes, plastics, hydrocarbon, antibiotic resistance, secondary metabolite synthesis and metal resistance from both the samples as studied indicate BoB to represent a highly diverse environmental niche for further exploration.

1. Introduction

Microbes are omnipresent in every environment and represent extreme diversity within members of archaea, bacteria and eukarya. As natural environment elicit variation in physio-chemical and atmospheric conditions, survival of microbes at any given environment depends on its adaptability and coordinative ability within a diverse community [1]. Oceans as a habitat, provides a complex, yet exceptional physical and chemical variability required for survival to all class of microbes, particularly marine bacteria [2]. The fact that even variations in marine bacterial diversity can occur within a few millimetres of ocean surface or sediments suggests the importance in understanding sediment microbial communities, their role in nutrient cycling and biological activities they perform for maintenance of marine biodiversity hotspots [3]. Accessing phylogenetic information at every level is a vital tool to profile microbial communities and compare microbial diversity across habitats [4]. Within marine habitat, deep-sea sediments represent an

extreme environment for microbial population due to high pressure, low temperature, deficiency in light, poor nutrient availability and are considered the most expansive, yet least studied environments [5]. Deep sea sediments are the largest reservoirs of organic carbon and microbial population present in deep-sea sediments play a vital role in carbon processing for maintenance of marine ecology at all energy levels [6]. Significant number of microbiological studies has been conducted on deep-sea sediments at various locations and have revealed the presence of phylogenetically diverse, metabolically active marine bacterial communities and the role they play in nutrient cycling through unique pathways [7–10].

Bay of Bengal, the largest bay in the world, is the discharge of major Indian rivers, Godavari, Krishna, Ganga, Brahmaputra, and Mahanadi. Sediment settlement at BoB is highly influenced by monsoonal rain and cyclonic storms across the mainland areas, where the rivers flow across and are documented to be rich in nutrient content [11]. With reference to investigations on deep sea and deep-sea sediments, BoB remains one

[☆] Metagenome data from this article have been deposited with the GenBank Database under Accession No SAMN12660966 and SAMN12660973

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of the less explored ecosystems [12]. Very limited studies have been performed to untap the microbial diversity of BoB sediment samples both at cultivable and metagenome levels [13–15]. Next generation sequencing technologies have enhanced the comprehensive analysis of microbial communities at deep sea sediments, wherein accessibility to samples remains a major setback [16]. However, analysis of microbial communities of deep-sea sediments at various taxonomic resolutions is important to understand the evolutionary, ecological and bioprospective significance of key microbes that are present which could be responsible for novel genes not reported so far [13].

High throughput studies have been undertaken for various oceanic locations [17–22] and 16S amplicon sequencing studies from BoB deep-sea sediments have reported microbial diversity at various depths [13,14]. The significant advantage of Nanopore sequencing technology is that the approach allowed improved taxa resolution compared to previous other technologies [23]. The higher information per read from nanopore sequencing is mainly useful as multiple genes present within one read could reveal functional potential of the study sites. High-throughput sequencing technologies have proven vital to examine microbial communities in deep-sea sediments in recent years [24]. However, despite such advancements, many ecosystems throughout the world remain yet to be investigated. The main objective of the study is to understand the microbial diversity, metabolic significance, ARG profile and bioremediation potential of deep-sea sediment samples from BoB through shotgun metagenome sequencing. To the best of our knowledge, microbial community and occurrence of ARGs, secondary metabolite genes and bioremediation potential from the BoB sediment at 3000 m depth has not been studied yet. The diverse bacterial community structure and key bacterial species identified from BoB sediment samples in the present study holds promise to further examine the biotechnological potential of functional genes they encode for industrial, biomedical and environmental applications.

2. Materials and methods

2.1. Sediment samples

Undisturbed and compact sediment samples were collected from the BoB employing a gravity corer at a depth of 3000 m during a cruise on Sagar Nidhi Research Vessel in 2017. Sampling location with coordinates is depicted in Fig. 1. Two samples, NIOT_S7 (sub-surface, 20–25 cm) and NIOT_S9 (deep layer, 55–60 cm) were separated from the sediment core and sediment properties were determined at SGS Laboratories, Chennai. The percentage of total organic carbon (TOC) was determined according to the method of Walkley and Black [25] (1934). Nitrogen and phosphate were determined according to Indian Standard 10,158:1982. Nutrient analysis of the sediment samples was carried out

using Skalar continuous flow analyser at National Center for Coastal Research (NCCR, Chennai). The concentrations of heavy metals, such as arsenic, mercury, selenium, and lead, were determined by the AOAC 990.08:2005 test method, and hexavalent chromium was determined by USEPA methods 3060A and 7196A.

2.2. DNA extraction, library preparation and nanopore sequencing

Total community DNA from sediment samples in triplicates (NIOT_S7 and NIOT_S9) were extracted using Qiagen DNeasy Power Soil kit (Qiagen, Hilden, Germany). As DNA yields were suboptimal, PCR enrichment was performed to amplify isolated DNA samples. Briefly, 2 µg of DNA from both the samples were end repaired (NEBnext ultra II end repair kit, New England Biolabs, MA, USA) and cleaned up with 1× AmPure beads (Beckmann Coulter, USA). PCR adapter ligation (PCA) was then performed with NEB blunt/TA ligase (New England Biolabs, MA, USA) and cleaned with 1× AmPure beads. Adapter ligated DNA samples were enriched using PCR reaction (LongAmp Taq 2× New England Biolabs, MA, USA) and cleaned up with 1× AmPure beads (Beckmann-Coulter, USA). The resulting enriched DNA were further used for library preparation using 1D Native Barcoding Genomic Kit (EXP-NBD103) and the library mix was cleaned up using 0.4× AmPure beads (Beckmann Coulter, USA) and finally eluted in 15 µl of elution buffer. The eluted DNA samples were subjected to sequencing on Grid-ION X5 (Oxford Nanopore Technologies, Oxford, UK) using SpotON flow cell (R9.4) in a 48 h sequencing protocol on MiniKNOW 2.1 v18.05.5 at Genotypic Technology, Bangalore.

2.3. Sequence processing and taxonomic analysis

The nanopore raw reads generated were basecalled and demultiplexed using Albacore (V2.1.10) [26]. Basecalled reads were error corrected using Canu [27] and were further used for gene prediction using MetaGeneMark [28]. The annotated genes were used to predict Gene Ontology, using Diamond blast-P [29] against Uniprot protein database. The identified genes were aligned against NR database using BLAST [30]. For taxonomic analysis and comparative studies, the NIOT_S7 and NIOT_S9 metagenomic datasets were compared to five 16S amplicon datasets available within BoB at different depths [13,14]. The reported five datasets were downloaded from the NCBI's SRA, uploaded into Metagenomic Rast Server MG-RAST (version 4.0.3) server [31], and processed for further analysis.

2.4. Gene annotation and gene ontology classification

The error corrected reads were used for gene prediction using MetaGeneMark [28] and Augustus [32]. The predicted ORFs were further

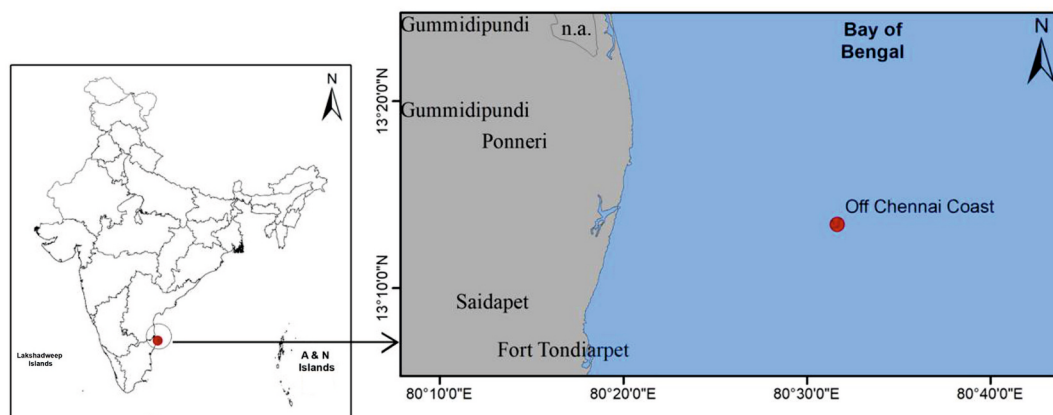


Fig. 1. Location of sampling site. Sediment sample was collected at a depth of 3000 m off Chennai coast in Bay of Bengal.

annotated using Diamond blast-P against Uniprot protein database (Bacteria, Archaea and Fungi). The Gene Ontology defines the concepts relating to gene functions ('GO terms'), and how these functions are related to each other ('relations'). The GO describes function with respect to three aspects: molecular function, cellular component, and biological process. Functional annotation was carried out to predict important metabolic pathway genes using KEGG Orthology (KOs) as key biomarkers in MG-RAST. The RPKM, (reads per kilobase per million sequenced reads, $RPKM = \text{mapped reads} / [\text{gene length in kb} \times \text{total reads (million)}]$) was examined to study the abundance of each gene as described by Zhang et al. (2018) [24].

2.5. Prediction of remediation related enzymes using RemeDB

Genes coding for enzymes that are involved in the breakdown of dye, hydrocarbon and plastic degradation were identified from metagenomic datasets of NIOT_S7 and NIOT_S9. RemeDB tool has been reported for rapid prediction of enzymes involved in bioremediation from high throughput metagenome datasets [33]. It consists of a well curated database of Pollutant Degrading Enzymes (PDE's) which uses Rapsearch [34] and Hmmer [35] to scan through metagenomic datasets for identifying signature sequences involved in bioremediation. RemeDB predicted the distribution of bioremediation enzymes from nanopore whole metagenome shotgun sequencing data of NIOT_S7 and NIOT_S9 sediment samples.

2.6. Annotation of ARGs, MRGs and mobile genetic elements

NanoARG, an online tool provides several features for detailed analyses of environmental derived metagenomic data obtained specifically by Nanopore sequencing. Profiling of ARG and MRE modules were performed in NanoARG using Deep-ARG LS model [36]. The deepARG-LS model is applied with permissive parameters, specifically, an identity cutoff of 25%, coverage of 40%, and a probability of 0.8, to predict that a region corresponds to an ARG. Abundance of ARGs was calculated based on the copy number of ARGs by an approach similar to that reported by Ma et al. [37]. The total hit reads of ARGs found within sediment metagenome samples were normalized to the size of the sequencing data as reported by Chen et al. [38].

2.7. Nucleotide sequence accession number

The base-called data sets were deposited as Fastq files in the NCBI SRA database with the following accession numbers: SAMN12660966 and SAMN12660973 and MG-RAST database with the following accession numbers: GT_SO_8247_S7 and GT_SO_8247_S9 (Project ID: mgp91045 and mgp91046).

3. Results and discussion

3.1. Distinct microbial communities within sub-surface and deep sediment layers

Microbial community composition analysis for sub-surface and deep surface samples from sediment core began with sediment sample characterization for NIOT_S7 and NIOT_S9 respectively. TOC of the sediment samples were found to be in the range of 0.85–0.89 (Table 1), indicating

a nutrient rich environment to support microbial communities. Heavy metal analysis in both the sediment samples revealed the presence of Ni, Co, Cu, Ag, Cd, Pb, Zn, Fe, Mg and total Chromium and their respective concentrations are shown in Table 2. Microbial community profiling with sediment samples were performed for which, DNA recovery from both the sediment samples was highly challenging due to the sediment organic content and humic acid inhibitor as reported by [39]. The recovered DNA concentrations were suboptimal for nanopore library preparation. Hence, PCR enrichment was performed to amplify whole metagenomic DNA for further analysis. Nanopore based sequencing of amplified sediment metagenome from NIOT_S7 and NIOT_S9 resulted in the generation of ~50 K quality-controlled reads from each sample. N50 range of 1500 to 2000 bp was observed for both sediment metagenomes and the read statistics are given in Supplementary Table S1. Within the total sequence reads, 12,968 sequences from NIOT_S7 and 13,801 sequences from NIOT_S9 were found to correspond to taxonomic signature sequences. Substantial numbers of sequences (~1000) within those assigned for microbial taxonomy were found to represent uncultured/unclassified taxa in both the samples.

Nineteen phyla were obtained from both the samples NIOT_S7 and NIOT_S9, with the most abundant being Proteobacteria (88%), followed by Bacteroidetes (3%), Firmicutes, Cyanobacteria and Actinobacteria (< 0.5%) in NIOT_S7 (Fig. 2a), whereas, NIOT_S9 was represented by Proteobacteria (82%), Bacteroidetes (12%), Firmicutes, Cyanobacteria and Actinobacteria (<1%). The most abundant class belonged to Alpha and Gammaproteobacteria which was evenly represented (46%) in NIOT_S7, whereas varied crucially (50% and 30%) in NIOT_S9. Other class such as Betaproteobacteria, Deltaproteobacteria, Flavobacteria, Bacilli and Actinobacteria also varied predominantly between samples NIOT_S7 and NIOT_S9 however, were represented at low levels (<1%). In concurrence with other previous studies, 99% of the total sequence reads generated in the present study accounted for bacteria [40,41]. Comparison to previous metagenomic studies on deep-sea sediments from BoB [14] and others showed that phylum Proteobacteria (82 to 88%) was found to be dominant in both the samples. Taxonomic diversity studies by Parvathi et al. from BOB sediment collected at a depth between 3000 and 3300 m also revealed that Proteobacteria accounted for 92.52% and remaining were represented by Firmicutes, Cyanobacteria, Bacteroidetes, Actinobacteria, Chloroflexi [14]. In contrast, taxonomic diversity analyses of BoB deep-sea sediment sample from a depth of 2000 m (NIOT.CHE.001) revealed that phyla Actinobacteria, was the dominant group [13] followed by Firmicutes, Proteobacteria and Chloroflexi. Reports from other marine environments including South China Sea [42] and North Atlantic [43] showed that most dominant microbial communities in these environments belong to phylum Proteobacteria. The phyla Bacteroidetes known to degrade high molecular weight compounds and biopolymers [44] were the second most abundant phyla in both the samples explored in this study. Firmicutes, often observed to be uniquely present in samples of deep-sea sediments collected at various locations was observed in both of the samples analysed [40,45].

At the order level, Rhodobacterales predominated in both the samples (55% and 49%). Alteromonadales (17%) and Thiotrichales (15%) were the second most observed order in NIOT_S7. Although Rhodobacterales were abundantly represented in NIOT_S9, Oceanospirillales (12%) and Cytophagales (12%) were the second most observed order in NIOT_S9 (supplementary fig. 1). Most of the families that are

Table 1
Nutrient composition of BoB sediment samples.

Samples	TOC	Total N	Total P	Nitrate	Nitrite	Ammonia	Silicate
NIOT_S7	0.58	464.36	9.03	203.82	1.54	212.1	71.05
NIOT_S9	0.55	308.84	9.29	164.88	1.75	126.84	91.98

TOC- Total Organic Carbon.

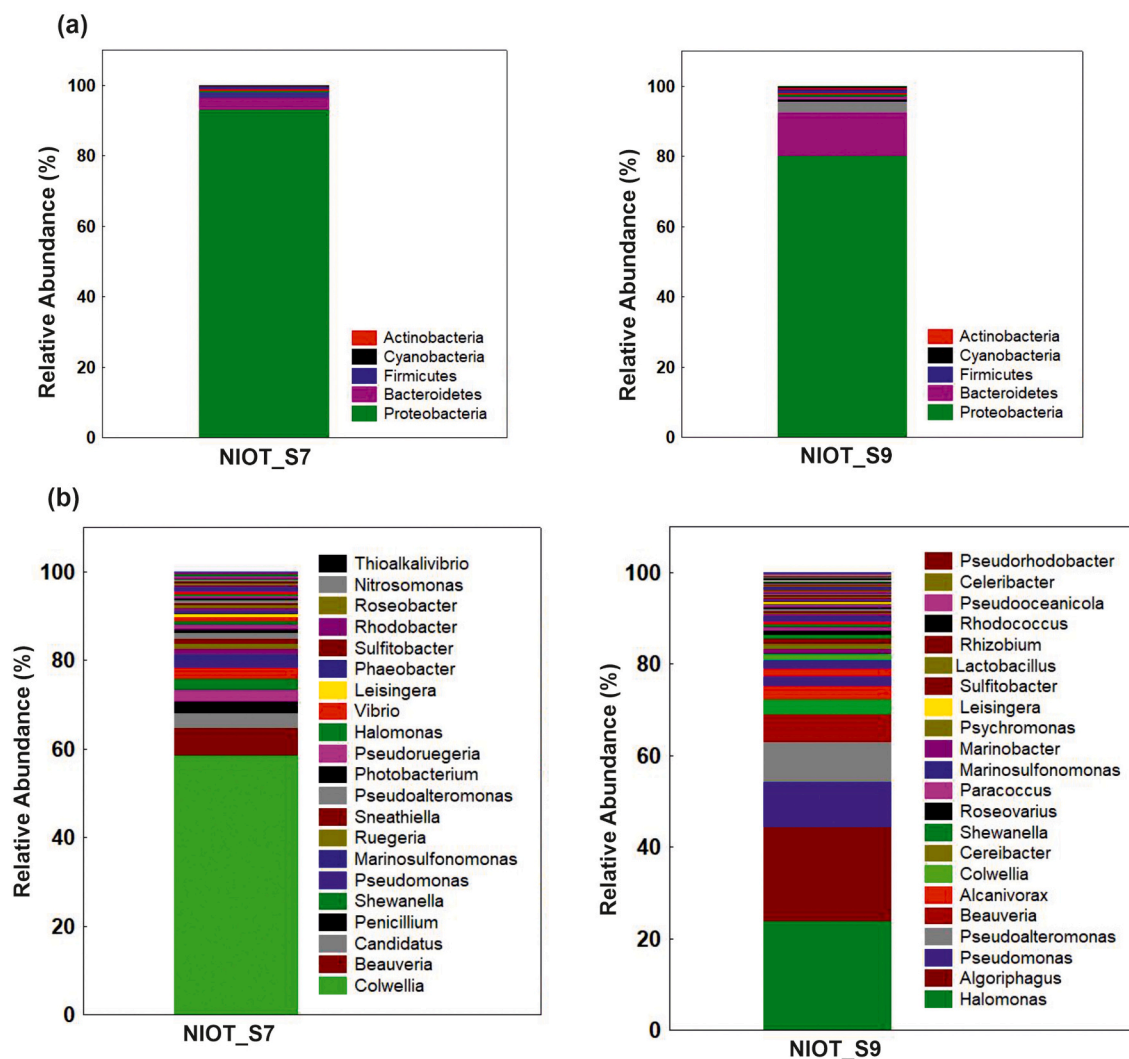
Values are represented as mg/kg.

Table 2

Concentrations of heavy metal ions from BoB deep-sea sediments.

Samples	Iron	Zinc	Cobalt	Copper	Lead	Manganese	Mercury	Nickel	Chromium
NIOT_S7	52,106.5	96.55	27.43	70	8.12	729.15	2.56	48.82	54.47
NIOT_S9	12,234.1	19.65	18.36	12.11	<0.1	1077.81	2.48	12.5	19.55

5 g of samples were used for the analysis. Values are represented in mg/kg of sample.

**Fig. 2.** Relative abundances of taxonomic distribution of NIOT_S7 and NIOT_S9 at the (A) phylum and (B) Genus level from BoB sediments.

predominant at top layer of sediment samples were suppressed at lower layer, and a trend in richness and abundance leading to enrichment of other microbial families was found increasing. Within Proteobacteria, Alphaproteobacteria and Gammaproteobacteria were found to be represented at identical levels in NIOT_S7. Notable differences in taxonomic tags corresponding to Alphaproteobacteria and Betaproteobacteria was observed with NIOT_S9. Alphaproteobacteria are oligotrophs, well known for their ability to survive in nutrient limiting conditions in environments such as deep-sea sediments [46]. Gammaproteobacteria are known to possess abilities to oxidize organic compounds for nutrients [47]. The dominant presence of Alpha and Gammaproteobacteria in both of the sediment samples analysed indicate the deep-sea sediment environment of BoB to have a complex nutrient rich environment, requiring extensive bacterial activity to process complex nutrients for microbial survival. The dominance of Alphaproteobacteria in both the samples reflected at bacterial order level, as Rhodobacterales was found

to be most abundantly represented by 50%. In contrast to other deep-sea sediment studies, from which Flavobacteriales were found to be the abundant order [48], Rhodobacterales was dominant in the present study and are shown to be responsible for biogeochemical cycling and redox reactions related to nutrient availability [49]. As observed with other deep-sea sediment samples, the rest of all other order were found to vary crucially in samples NIOT_S7 and NIOT_S9 [50–52]. Colwelliaceae, Rhodobacteriaceae and Thiotrichaceae are predominantly represented in NIOT_S7. Members of Halomonadaceae and Cyclobacteriaceae which is known to possess antibiotic resistance genes [53] were also found to be represented in notable levels in NIOT_S9. All genera having above 1% abundance in both the samples were screened to obtain the dominant genera and interestingly no common pattern was found to be present at both the locations. Our analysis at genus level with sequences from NIOT_S7 and NIOT_S9 indicated *Colwellia* (58%) to be dominant in NIOT_S7, whereas, *Halomonas* (23%), *Algoriphagus* (20%) and

Pseudomonas (9%) were prominent members in NIOT_S9. Except for above mentioned genera, >90% of all other genus in both the samples were represented by <1% of sequences (Fig. 2b). Visualization of the dominant species diversity and abundance showed clear difference, albeit with few similarities, for each sample (Supplementary fig. 2).

Consequently, in the sediment sample NIOT_S7, dominance of *Thiotrichaceae* bacterium, *Rhodobacterales* bacterium and *Colwelliapsychrerythrae* was observed at high percentage levels (36–9%). Bacteria belonging to *Thiotrichaceae* family exhibit functional capacities for nutrient cycling including nitrogen and sulfur. The strictly psychrophilic genus *Colwellia* from Gammaproteobacteria are important for carbon and nutrient cycling. *Rhodobacterales* belonging to alphaproteobacteria are non-sulfur bacteria that have the ability to fix nitrogen was the only predominant genus in bottom layer sample NIOT_S9. Genus *Clowellia*, *Marino sulfonomonas*, *Sneathiella*, *Pseudomonas*, *Kangiella* and *Shewanella* being represented at notable levels in NIOT_S7 correlates with the high complex nutrient availability wherein probably specific metabolic pathways and higher extracellular protein secretion by these microorganisms may render them with the ability for survival. *Algoriphagus*, *Halomonas*, *Cereibacter* and *Pseudoalteromonas* were found to be dominant in lower layer sample NIOT_S9. Members of these genera observed in lower layer sample are known for bioactive compounds production with ability to remain dormant in nutrient limited conditions. In NIOT_S9, except *Rhodobacterales* bacterium (29%) all other species were represented by <5% of sequences.

In order to perform comparative studies on bacterial diversity and sediment properties of NIOT_S7 and NIOT_S9 with other similar ecological samples, 16S data sets from similar sites [13,14] at different depths were downloaded from NCBI database. The results of diversity analyses revealed completely contrast and distinct microbial profile. To further analyse and visualize the correlation between the bacterial communities and the nutrient properties of the sediments, a canonical correspondence analysis was performed. The plot showed that nutrient properties of sediment samples with reference to TOC, TP and TN were represented in different quadrants (Fig. 3). Bacterial classes such as Alpha, Beta and Gammaproteobacteria, on the positive side of axis 1 are

influenced by the depth of the sediment in NIOT_S7 and NIOT_S9. Actinobacteria and Bacilli are positively correlated with TP and TN which influences NIOT.CUD.001 and NIOT.CHE.001 community [13]. Sub-sea floor sediments exhibit distinct microbial communities based on environmental properties and nature of sediment as reported [54,55].

3.2. Ecological significance of deep-sea sediment communities

Though the metabolic potential has been analysed in many recent studies such as those from the South Atlantic polar front, South China sea, Eastern Gulf of Mexico, Southwest Indian ridge, Sao Paulo Sea, Great Australian Bight [41,42,52,56–58], the functional significance of microbial community in the deep-sediment of BoB has not been studied in detail. To gain insight into the metabolic potential of microbial communities between sediment layers, relative abundance of key microbial metabolic genes was analysed. Signature genes corresponding to carbon, nitrogen and sulfur pathways were found to be represented in both NIOT_S7 and NIOT_S9 data sets with difference in their relative abundance. The functional groups and pathway modules corresponding to each data set were identified using KEGG annotation (Supplementary Table S2). A total of ~one million genes were identified in samples NIOT_S7 and NIOT_S9. Gene targeted analyses were performed by examining metagenomic data and it was found that the metabolic genes present in bacterial communities were mainly involved in carbohydrate and amino acid metabolism. Genes involved in carbon acquisition and breakdown including carbohydrate active enzymes (CAZY family) as well as transporters were observed in both annotated data sets. Chitinase and *N*-acetyl-glucosaminidase genes involved in breakdown of chitin, the principal structural component of algal, fungal cell walls and exoskeleton of arthropods were annotated from Cytophagia belonging to Bacteroidetes phylum from NIOT_S9. The carbon fixation metabolism from NIOT_S7 and NIOT_S9 community was investigated based on the annotation of key enzymes involved in the reductive acetyl-CoA pathway. Carbon monoxide dehydrogenase was predicted in both NIOT_S7 and NIOT_S9 and was found to belong to Alphaproteobacteria and Deltaproteobacteria, respectively. Previous report suggests that

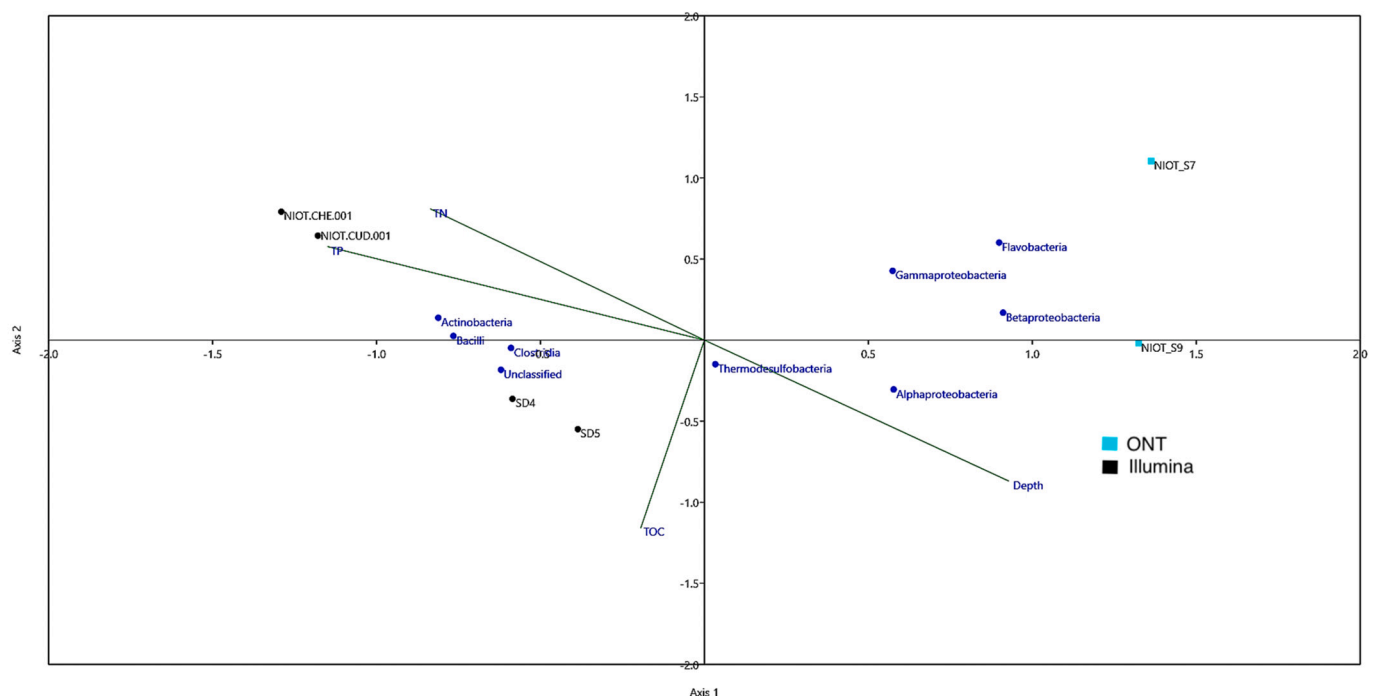


Fig. 3. Canonical Correspondence Analysis (CCA) plot for major bacterial class (comprising >0.5% of the total bacterial community composition of each sample) from BoB deep-sediment datasets. NIOT.CHE.001, NIOT.CUD.001, SD4, SD5 and SD6 are previous studies from Bay of Bengal at various depths [13,14]. ONT- Oxford Nanopore Technology.

acetate, lactate and butyrate fuel the respiratory pathway in sediment community [59], and similarly sequences that would represent lactate, alcohol and aldehyde dehydrogenases were annotated in NIOT_S7 and NIOT_S9 (Supplementary Table S3). Protein breakdown by peptidases play a vital role as proteins are the main source of bioavailable carbon and nitrogen for the sediment inhabiting microbial community [59]. Sequences that were annotated as peptidase coding genes with signal peptide sequence, metallo peptidases from M16 (metalloendopeptidase), M23 (beta-lytic metallopeptidase- known to lyse cells of other bacteria as defensive or feeding mechanism) and M48 (Ste24 endopeptidase) families were predominantly identified from Proteobacteria and Bacteroidetes in NIOT_S7 and NIOT_S9. Peptidase S8 from subtilisin family was abundant from NIOT_S7 as compared to NIOT_S9 (Supplementary Table S4).

KEGG annotation revealed respiratory pathway genes in both the data sets. Genes for assimilatory sulfate reduction pathway in NIOT_S7

metagenome have been annotated predominantly from Gamma and Betaproteobacteria (Supplementary Table S5). Dissimilatory sulfite reductase, *dsrAB* sequences in the datasets were mainly assigned to Gammaproteobacteria (Chromatiales and Thiotrichales) and adenosine-5'-phosphosulfate reductase, *aprAB* sequences from Gammaproteobacteria (Chromatiales, Acidiferrobacterales) and Betaproteobacteria (Nitrosomonadales). Sulfatases that hydrolyze sulfur esters and aryl sulphates may account for acquisition of sulfur for growth or energy production. In the present study, *sat* (sulfate adenylyl transferase) genes of the dissimilatory sulfate reduction pathway were assigned to Gammaproteobacteria (Alteromonadales) and their abundance is shown in Fig. 4a for NIOT_S7 and NIOT_S9, respectively. Interestingly, genes responsible for inorganic sulfur transformations like *sudA* (sulfide dehydrogenase), *psrA* (polysulfide reductase), *ttrA* (tetrathionate reductase) and *rdlA* (rhodanese-like protein) were annotated from NIOT_S7 whereas only *ttrA* was annotated from NIOT_S9.

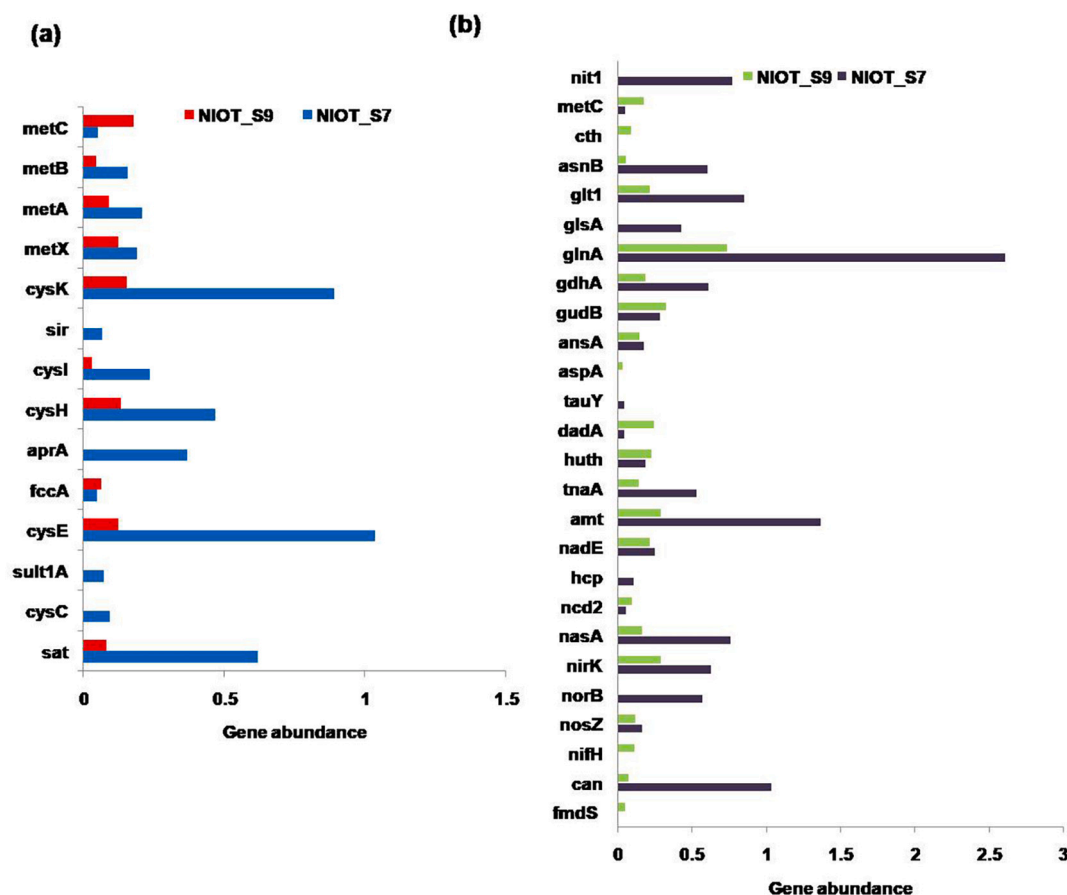


Fig. 4. Gene abundances involved in bacterial sulfur and nitrogen pathways from both the metagenomes. (a) Gene abundances involved in the microbial sulfur pathways in the metagenome data sets of BoB deep-sea sediment. *cysE*, serine *O*-acetyltransferase; *cysH*, phosphoadenosinephosphosulfate reductase; *cysI*, sulfite reductase (NADPH) hemoprotein beta-component; *sir*, sulfite reductase(ferredoxin); *cysK*, cysteine synthase; *metX*, homoserine *O*-acetyltransferase/*O*-succinyltransferase; *metA*, homoserine *O*-succinyltransferase/*O*-acetyltransferase; *metB*, cystathionine gamma-synthase; *metC*, cysteine-S-conjugate beta-lyase; *soxA*, sulfur oxidizing protein; *soxB*, thiosulfohydrolase; *soxC*, sulfite dehydrogenase; *soxD*, sulfur dehydrogenase; *fccA*, flavocytochromeC/sulfite dehydrogenase; *fccB*, flavocytochromeC/sulfite dehydrogenase; *atsA*, aryl sulfohydrolase; *suyA*, sulfolactatesulfo-lyase; *suyB*, sulfolactatesulfo-lyase; *tauD*, taurine dioxxygenase; *xsc*, sulfoacetaldehyde acetyltransferase; *psrA*, polysulfide reductase; *ttrA*, tetrathionate reductase; *sudA*, sulfide dehydrogenase; *rdlA*, rhodanase. (b) Methane pathway genes are represented as *asexA*, alcohol dehydrogenase; *catB*, catalase; *qhpA*, quinohemoprotein amine dehydrogenase; *gfa*, S-(hydroxymethyl)glutathione synthase; *adhC*, S-(hydroxymethyl)glutathione dehydrogenase / alcohol dehydrogenase; *frmB*, S-formylglutathione hydrolase; *fdoG*, formate dehydrogenase major subunit; *cooS*, anaerobic carbon-monoxide dehydrogenase catalytic subunit; *hoxH*, NAD-reducing hydrogenase large subunit; *tmuM*, 1,3,7-trimethyluric acid 5-monooxygenase; *dmd/tmd*, dimethylamine/trimethylamine dehydrogenase. (b) Gene abundances involved in the microbial nitrogen pathway in the metagenome datasets of BoB deep-sea sediment. *fmdS*, formamidase; *can*, carbonic anhydrase; *nifH*, nitrogenase iron protein; *nosZ*, nitrous-oxide reductase; *norB*, nitric oxide reductase subunit B; *nirK*, nitrite reductase (NO-forming); *nasA*, assimilatory nitrate reductase catalytic subunit; *ncd2*, nitronate monooxygenase; *hcp*, hydroxylamine reductase; *nadE*, NAD⁺ synthase; *amt*, aminomethyltransferase; *tnaA*, tryptophanase; *huth*, histidine ammonia-lyase; *dadA*, D-amino-acid dehydrogenase; *tauY*, taurine dehydrogenase large subunit; *aspA*, aspartate ammonia-lyase; *ansA*, L-asparaginase; *gudB*, glutamate dehydrogenase; *gdhA*, glutamate dehydrogenase; *glnA*, glutamine synthetase; *glsA*, glutaminase; *glt1*, glutamate synthase (NADH); *asnB*, asparagine synthase; *cth*, cystathionine gamma-lyase; *metC*, cysteine-S-conjugate beta-lyase; *nit*, nitrilase.

Sulfide quinone oxidoreductase (*sqr*) and flavocytochrome-C sulfide dehydrogenase (*fccAB*) encode enzymes that mediate the oxidation of sulfide to polysulfide and were annotated in NIOT_S7, however *sqr* was absent in NIOT_S9. Genes that constitute thiosulfate oxidizing multienzyme system of the SOX pathway, such as *soxABXYZ* and *soxCD* oxidize various reduced sulfur compounds to sulphate [60] and are represented by sequences of Alpha and Gammaproteobacteria in NIOT_S7, but only by Alphaproteobacteria in NIOT_S9. Also, genes related to organic sulfur transformation, *atsA* (arylsulfatase), *suyA*, *suyB*, (sulfolactatesulfo-lyase) and *xsc* (sulfo acetaldehyde acetyltransferase) from Alphaproteobacteria and Bacteroidetes were annotated only from NIOT_S7 metagenome. Overall, results obtained suggest that microbial communities from NIOT_S7 play a crucial role in assimilatory and dissimilatory sulfate reduction pathways. It has been shown that the sulfur cycle of marine sediments is mainly driven by dissimilatory sulfate reduction pathway of anaerobic microorganisms [61]. Key enzymes of dissimilatory sulfate reduction are used to analyse the distribution and abundance of sulfate reducing microorganisms [62]. Signature genes such as nitrite reductase, nitrate reductase of nitrogen pathway was determined and they were found to be represented in both data sets with difference in their relative abundance. The functional potential of the community is distinct with sulfur and nitrogen pathways and were dominant in NIOT_S7 in comparison to NIOT_S9. Majority of the key genes observed in the datasets belong to families Colwelliaceae, Thiotrichaceae within the class Gammaproteobacteria implying their role in sulfur cycle. It has been reported that Epsilonproteobacteria uses SOX pathway for sulfur oxidation in deep-sea hydrothermal fields [63]. In our report, Alphaproteobacteria dominate the SOX oxidation pathway in NIOT_S7 whereas sulfur oxidizing Alphaproteobacteria, Rhodobacteria and Bacteroidetes has been reported from hydrothermal vent in SWIR and Chimneys of Lau Basin [64].

Key functional genes involved in nitrogen cycle (nitrogen fixation, dissimilatory nitrate reduction and denitrification) were investigated in both the metagenomes (Supplementary Table S6; Fig. S3). NIOT_S7 showed higher abundance at a level of three-fold higher for *glnA* which is involved in ammonia assimilation. Additionally, *norB*, *nirK* and *nosZ* genes encoding the enzymes for nitrate, nitrite reductases and nitrous oxide reductases were abundant in NIOT_S7 as compared to NIOT_S9 (Fig. 4b). The gene abundance for assimilatory nitrate reductase (*nasA*) from NIOT_S7 was four-fold higher than NIOT_S9. Nitrogenase (*nifH*) gene involved in nitrate reduction to ammonia was identified in NIOT_S7 whereas it was absent in NIOT_S9. Nitrogen transformation mediated by sediment microbes is crucial as the pathway links the organic matter degradation and nutrient regeneration [65]. Key genes of the dissimilatory nitrate reduction and denitrification pathways were found to be predominating in NIOT_S7 dataset. The respective taxa mainly belong to Alpha and Gammaproteobacteria. Overall, the metabolic potential with respect to sulfur and nitrogen pathways was mainly observed in NIOT_S7. Genes related to dissimilatory sulfate reduction and sulfur oxidation were detected in sediment metagenomes of Yap Trench [24]. Metabolic prediction of bacterial communities from three deep-sea sediment of BoB revealed that sulfide oxidizers are mainly involved in sulfur metabolism [14]. Taxa involved in sulfate reduction are reported from Barren Island sediment sample NIOT.BAR.001 [13].

3.3. Identification of genes involved in bioremediation and secondary metabolism

Biodegradation of hydrocarbons is an important biogeochemical process in a variety of deep-sea environments [66,67]. Studies have shown the ability of anaerobic microorganisms to metabolize hydrocarbons, and mono- and polycyclic aromatic compounds [68]. With an aim to harness novel genes that could aid bioremediation of dyes, plastics and aromatic hydrocarbons, metagenomic data sets of NIOT_S7 and NIOT_S9 were explored for those gene sequences employing REMEDB. Results obtained from REMEDB search indicated 37, 29 and

51 sequences related to dye, hydrocarbon and plastic enzyme signatures from NIOT_S7 and 9, 15 and 18 reads of respective genes from NIOT_S9 (Fig. 5). Phenylacetaldehyde hydrogenase enzyme involved in plastic degradation was found to be abundant in NIOT_S7. Similarly, methyl-esterase and arylsulfatase enzyme involved in hydrocarbon and dye degradation respectively were found to be enriched in NIOT_S7 metagenome in comparison to NIOT_S9. Overall, the number of enzymes predicted in metagenomic datasets was abundant in NIOT_S7 as compared to NIOT_S9. The genes coding for styrene monooxygenase, styrene-oxide isomerase and phenyl acetaldehyde dehydrogenase involved in the degradation of polystyrene was observed mainly from NIOT_S7. The number of reads for methyl-esterase gene, which play a role in alkane degradation [69] was higher in NIOT_S7 than NIOT_S9. Plastic and hydrocarbon degradation genes observed signifies the potential of microbial community in NIOT_S7 to exhibit bioremediation significance.

Deep-sea microbes have unique metabolic capabilities that help their survival in the habitat by influence of their secondary metabolites production capability [70]. Such secondary metabolites from microbial resources have been explored as biotechnological tools for application in sustainable industrial process, agriculture and medicine [71]. Polyketide and nonribosomal peptide producing gene clusters have not been studied in detail from marine sediments. KEGG annotation for secondary metabolite pathways (Supplementary Table 2) revealed that reads relevant to biosynthesis of secondary metabolites and terpenoids were enriched in NIOT_S7. Reads related to indole and isoquinoline alkaloids and siderophore group of nonribosomal peptides were annotated in both the data sets. Apart from polyketide synthases and non-ribosomal polyketide synthases, several secondary metabolite genes were annotated majorly in NIOT_S7. In an attempt to identify full length type III polyketide synthase genes, BLASTX analyses of metagenomic sequences with in-house created PKSdb (https://www.niot.res.in/niot1/mbt_activities.php) was performed. Reads related to Type III PKS genes were annotated from NIOT_S9 data. A full-length type III PKS sequence (39% identity to putative type III PKS from *Algoriphagus halophilus*) was identified from 5 kb sequence derived from NIOT_S9 (Table 3). Recently, the occurrence and diversity of PKS I and NRPKS gene clusters have been studied from Yellow Sea sediment through metagenomic approach [72]. In this study, a type III polyketide synthase gene was identified and annotated against in-house PKS database. Full length PKS predicted from this study helped to understand the significance of long read technology and to explore further the functional potential of BOB sediment.

3.4. Abundance of ARGs and MRGs within sediment layer

The presence of antibiotic resistance genes from BoB sediment ecosystems has not been explored previously. NanoARG revealed the presence of antibiotic resistance genes from both the samples. The total hit reads for ARGs found within sediment metagenome samples were normalized to the size of the sequencing data as reported by Chen et al. (2013) [73]. Genes responsible for multidrug resistance (MDR), aminoglycoside, glycopeptides, macrolide lincosamide-streptogramin (MLS) and beta-lactam antibiotics were annotated in NIOT_S7 and NIOT_S9 (Fig. 6a). MDR was the most abundant of all ARGs in NIOT_S7 and suggested the presence of MDR efflux pumps among the microbial communities of sub-surface sediment sample. Aminoglycoside resistance genes were abundant in NIOT_S9, whereas fosmidomycin and triclosan were exclusively observed in NIOT_S7. The ancient history of antibiotic resistance is evidenced by the discovery of genes coding for β -lactam and tetracycline resistance in 30,000-year-old permafrost sediments [74]. In natural environments, competition between microbes has been proved to be the driving tool for evolution of resistance genes [75]. Studies have been performed in the Indian Gulfs, which reported the occurrence and distribution of ARGs and corresponding microbial community [76]. Some of the recent studies from other locations had

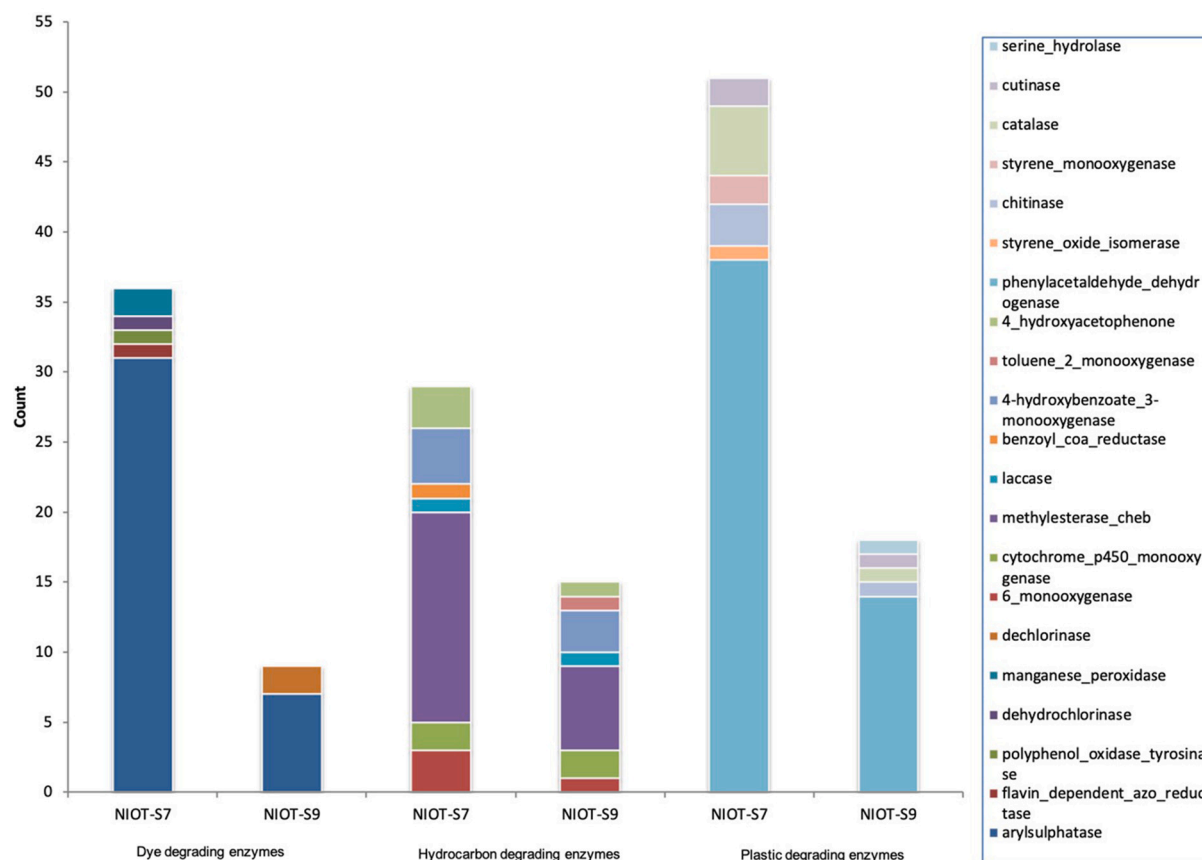


Fig. 5. Total number of counts from NIOT_S7 and NIOT_S9 involved in bioremediation (dye, plastic and aromatic hydrocarbon degradation) were annotated using REMEDB.

Table 3

Identification of putative polyketide synthase genes from metagenome data sets.

	Query id	Subject id	Gene	E value	% identity
NIOT_S9	74c5b36b-469d-4 cc2-9fa1-b9ba2be34d74	WP_074226094.1 <i>Algoriphagus</i>	type III polyketide synthase	3.84E ⁻³⁰	39.16
	93ce73cc-81b8-4119-ab35-eba51752ee0b	WP_131595860.1 <i>Pedobacter</i> sp.	type III polyketide synthase	1.75E ⁻⁴	36.0
	0cee7258-3f3f-4387-b29c-d3769eb1249b	WP_001100525.1 <i>Staphylococcus</i> sp.	ketoacyl-ACP synthase III	1.86E ⁻⁸	37.8
	795dd294-1569-4488-a594-b660aba6db08	WP_001100528.1 <i>Staphylococcus</i>	ketoacyl-ACP synthase III	9.22E ⁻²¹	41.8
	b1738b62-e2c6-415d-af47-30749afd41f6	WP_005478983.1 <i>Streptomyces bottropensis</i>	oxoacyl-ACP synthase III family protein	0.003	31.0
NIOT_S7	273b9ff5-facc-4ea3-bb32-ba949be4fe5d	WP_131595860.1 <i>Pedobacter</i> sp.	type III polyketide synthase	0.004	31.3
	4035f33b-32f2-42fe-ae85-6233a8148264	WP_089759886.1 <i>Chitinophaga terrae</i>	type III polyketide synthase	2.21E ⁻⁵	31.6
	011d6d77-5360-424c-b5f5-2574e300e44f	WP_087452578.1 <i>Ralstonia solanacearum</i>	hybrid non-ribosomal peptide synthetase/type I polyketide synthase	4.73E ⁻³³	39.4
	795dd294-1569-4488-a594-b660aba6db08	WP_001100526.1 <i>Bacillales</i>	ketoacyl-ACP synthase III	8.49E ⁻³³	41.7

reported the presence of antimicrobial resistance genes from deep sea sediments and sea water [73,77]. D'Costa et al. have reported the diverse and abundant nature of ARGs in Beringian permafrost sediments [74]. It is intriguing to observe the presence and type of ARGs from BoB sediment layers. The occurrence of multidrug resistance genes and other ARGs showed that the presence of antibiotic resistance in sediment community is natural. Pelagic sediments in the Gulf of Khambat

revealed 2354 unique types of resistance genes that mainly belongs to macrolides, glycopeptides and peptide antibiotics [75]. In coastal sediments of East China Sea, sulfonamides and tetracycline resistance genes were the most prevailing types of ARG [38]. In Antarctic soil, main ARG type was bacitracin, trimethoprim and chloramphenicol resistance genes whereas from deep sea sediments trimethoprim, MLS, MDR and bacitracin were observed [78]. In South China Sea sediments, the most

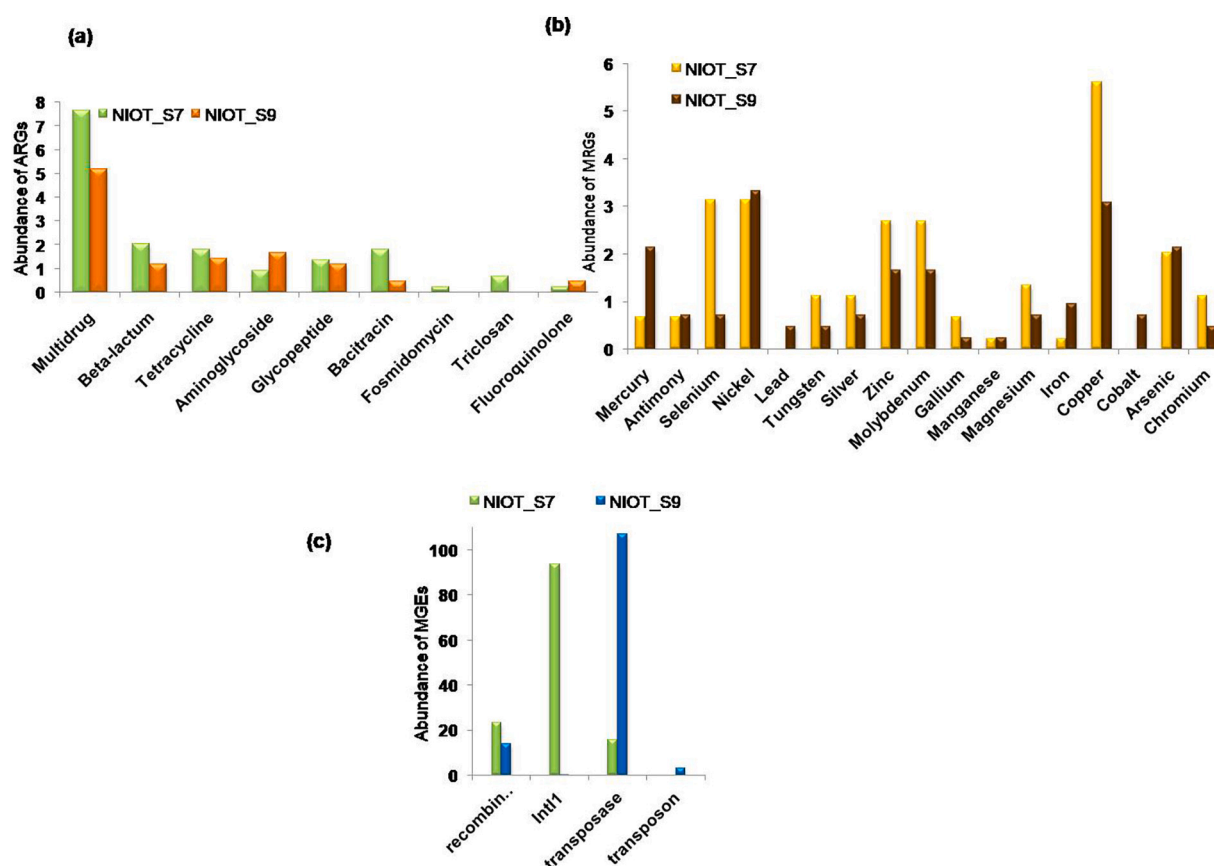


Fig. 6. Relative abundance of ARGs, MRGs and MGEs in NIOT_S7 and NIOT_S9 sediments. (a) antibiotic resistance classes (b) metal resistance genes (c) mobile genetic elements.

prevalent and abundant ARGs are those related to macrolides and polypeptides resistance [29]. ARGs were present at lower levels in the polar marine sediments [79]. Similarly in our study, the overall abundance of ARGs was less. Along with ARGs, genes involved in arsenic and metal resistance were also annotated from both the datasets. The presence of mobile genetic elements in BoB metagenome sequence signifies inhabiting microbes could have the ability to transfer ARG clusters.

Heavy metal concentrations determined from NIOT_S7 and NIOT_S9 revealed that NIOT_S7 exhibited higher concentration of iron, copper, zinc, cobalt, lead, nickel and chromium when compared to NIOT_S9 (Table 1). Manganese concentration was higher in NIOT_S9. Selenium, molybdenum and copper resistance genes were abundantly represented in NIOT_S7 which correlates with higher concentration of copper from the sample. Metal resistance and antibiotic resistance are associated due to co and cross-resistance mechanisms [80]. In addition to antibiotic resistance genes, sediment metagenomes of BoB harboured genes involved in resistance to heavy metals such as mercury, antimony, selenium, nickel, tungsten, silver, zinc, molybdenum, gallium, manganese, magnesium, iron, copper, arsenic and chromium respectively (Fig. 6b). Arsenic and nickel resistance genes were equally abundant in NIOT_S7 and NIOT_S9 whereas mercury resistance was twofold higher in NIOT_S9. Microbial communities at the deep-sea environment survive in metal rich conditions based on their metal resistance mechanism.

4. Conclusion

A metagenomic snapshot of the BoB sediment ecosystem was performed to understand the signatures of microbial community and their functional significance. An initial study on microbial communities from deep-sea sediment of BoB has been revealed and a roadmap for mining novel gene such as those for type III polyketide synthase from the

metagenome data has been laid through the long read sequencing approach. Thus, the study may be a significant repository of novel genes with potential significance. The results of the study reveal a distinct microbial ecology with metabolic potential. The results could be potentially used in several downstream studies such as comparative analysis of deep-sea metagenomes from BoB, Andaman Sea and Southern Indian Ocean. Analysis of multiple and different deep-sea ecosystems are necessary to establish a comprehensive understanding of diversity and function of microbial communities inhabiting these environments. Further, analysis of metabolic versatility of microbial community in the deep-sea sediment of BoB could reveal newer gene targets and possibility of bioprospection.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ygeno.2022.110524>.

Authors' contributions

Conceptualization, funding acquisition, and project management: MJ. VR collected the samples and VR, BK, SS, VD performed computational data analysis. Data curation: MJ, VR, BK, SS, VD, NTK. Manuscript writing and editing: MJ, VR, BK, SS, VD, NTK, JKL, RK, DG; All of the authors contributed to drafting the manuscript and approved the final version.

Funding

The project was supported by a grant from Ramalingaswami Fellowship (BT/RLF/Re-entry/56/2012) from the Department of Biotechnology, Ministry of Science and Technology, New Delhi to M.J.

Declaration of Competing Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

Data will be made available on request.

Acknowledgements

We greatly acknowledge NIOT, Ministry of Earth Sciences, Government of India, for research facility and permitting sample access. The analyses of nutrients in sediment samples were performed at NCCR, Chennai and their support is greatly acknowledged.

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